

Application Note

Bare-metal Boot Code for ARMv8-A Processors

Version 1.0

Non-Confidential



Bare-metal Boot Code for ARMv8-A Processors

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Release Information

The following changes have been made to this Application Note.

Document History			
Date	Issue	Confidentiality	Change
31/03/2017	A	Non-Confidential	First release

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LES-PRE-20349

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1 Conventions and Feedback

The following section describes the typographical conventions and how to give feedback:

Typographical conventions

The following typographical conventions are used:

`monospace` denotes text that can be entered at the keyboard, such as commands, file and program names, and source code.

`monospace` denotes a permitted abbreviation for a command or option. The underlined text can be entered instead of the full command or option name.

monospace italic denotes arguments to commands and functions where the argument is to be replaced by a specific value.

monospace bold denotes language keywords when used outside example code.

italic highlights important notes, introduces special terminology, denotes internal cross-references, and citations.

bold highlights interface elements, such as menu names. Also used for emphasis in descriptive lists, where appropriate, and for ARM® processor signal names.

Feedback on documentation

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- The title.
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- ARM Glossary, <http://infocenter.arm.com/help/topic/com.arm.doc.aeg0014-/index.html>.

2 Preface

This preface contains the following topics:

- *References* on page 8.
- *Terms and abbreviations* on page 9.

2.1 References

- *ARM® Architecture Reference Manual ARMv8, for ARMv8-A architecture profile* (ARM DDI 0487).
- *ARM® Cortex™-A Series Programmer's Guide for ARMv7-A* (ARM DEN 0013).
- *ARM® Cortex®-A Series Programmer's Guide for ARMv8-A* (ARM DEN0024).

2.2 Terms and abbreviations

Abbreviations and terms used in this document are defined here.

EL	Exception level.
MMU	Memory Management Unit.
PL	Privilege Level.
SoC	System on Chip.
SP	Stack Pointer.
TRM	Technical Reference Manual.

3 Introduction

This chapter describes the purpose and scope of this application note.

It contains the following topics:

- *Document purpose* on page 11.
- *Document scope* on page 12.

3.1 Document purpose

Hardware verification engineers often run bare-metal tests to verify core-related function in a *System on Chip* (SoC). However, it can be challenging to write boot code for a bare-metal system, without a basic understanding of software development on the ARM architecture.

This application note assumes that you are not familiar with ARM software development. It is intended to help you write boot code for ARMv8-A processors.

You can reference the boot code examples in this application note, and write your own boot code for a bare-metal system that is based on ARMv8-A processors.

3.2 Document scope

This application note provides code examples for the following important operations that are involved in booting a bare-metal system:

- Initializing exceptions.
- Initializing registers.
- Configuring the MMU and caches.
- Enabling NEON and Floating Point.
- Changing Exception levels.

The code examples are written with the GNU assembly grammar and are tested on the Cortex-A53, Cortex-A72, and Cortex-A73 processors. They also apply to other ARMv8-A processors.

The ARMv8-A architecture supports two different Execution states:

- AArch32.
- AArch64.

This application note provides boot code examples for each Execution state.

For boot code examples applicable to ARMv7-A processors, see the *ARM® Cortex™-A Series Programmer's Guide* for ARMv7-A.

4 Boot code for AArch32 mode

Read this chapter for boot code examples for AArch32.

It contains the following topics:

- *Initializing exceptions on page 14.*
- *Initializing registers on page 16.*
- *Configuring the MMU and Caches on page 21.*
- *Enabling NEON and Floating Point on page 28.*
- *Changing modes on page 30.*

4.1 Initializing exceptions

Exception initialization requires setting up the vector tables and enabling asynchronous exceptions.

4.1.1 Setting up a vector table

When booting a processor in AArch32 mode, the value of SCTLR.V sets the location of the reset vector:

- When SCTLR.V is 0, the processor starts execution at address 0x00000000.
- When SCTLR.V is 1, the processor starts execution at address 0xFFFF0000.

You can use the hardware input **VINITHI** to set the reset value of SCTLR.V.

For exceptions other than reset, the processor looks up vector tables, which can be placed at customized places by programming vector base address registers. There are up to four vector tables. The corresponding vector base address registers are:

- *Vector Base Address Register (VBAR) (Secure).*
- *Monitor Vector Base Address Register (MVBAR).*
- *Hyp Vector Base Address Register (HVBAR).*
- VBAR (Non-secure).

Example 4-1 shows a typical vector table that is used for reset and other exceptions.

Example 4-1 Typical vector table

```
.balign 0x20
vector_table_base_address:
B   reset_handler
B   undefined_handler
B   svc_handler
B   prefetch_handler
B   data_handler
NOP
B   IRQ_handler
// You can place the FIQ handler code here.
```

The vector entries in the four tables might be different. For details, see the section, *Exception vectors and the exception base address*, in the *ARM® Architecture Reference Manual ARMv8, for ARMv8-A architecture profile*.

You must initialize the four vector tables, and program the vector table base address registers before using the vector tables. The base addresses of vector tables must be 32-byte aligned.

Example 4-2 shows you how to initialize VBAR and MVBAR after reset.

Example 4-2 VBAR and MVBAR initialization

```
LDR    R1, =secure_vector_table_base_address
MCR    P15, 0, R1, C12, C0, 0                // Initialize VBAR (Secure).

LDR    R1, =monitor_vector_table_base_address
MCR    P15, 0, R1, C12, C0, 1                // Initialize MVBAR.
```

4.1.2 Enabling asynchronous exceptions

Asynchronous exceptions include asynchronous abort, IRQ and FIQ. They can be masked by CPSR.{A,I,F} register bits after reset. Therefore, if asynchronous aborts, IRQ and FIQ are to be taken, the CPSR.{A,I,F} bits must be cleared.

To enable interrupts, you must also initialize the external interrupt controller to deliver the interrupt to the processor, but it is not covered in this document.

Example 4-3 shows you how to enable asynchronous abort, IRQ and FIQ.

Example 4-3 Asynchronous abort, IRQ and FIQ exceptions enablement

```
// Enable asynchronous aborts, interrupts, and fast interrupts.
CPSIE    aif
```

4.2 Initializing registers

Register initialization involves initializing the following registers:

- General purpose registers.
- Stack pointer registers.
- System control registers.

4.2.1 Initializing general purpose registers

Some registers in ARM processors use non-reset flip-flops. This can cause X-propagation issues in hardware simulations. Register initialization reduces the possibility of this issue.

————— Note —————

This initialization is not required on silicon chips because X status only exists in hardware simulations.

Example 4-4 shows you how to initialize general-purpose registers after reset. Because there are banked general-purpose registers for different modes in AArch32, the example code changes to different modes and initializes them all.

Example 4-4 General-purpose registers initialization

```
// Processors are in Secure SVC mode after reset.
MOV    R0, #0
MOV    R1, #0
MOV    R2, #0
MOV    R3, #0
MOV    R4, #0
MOV    R5, #0
MOV    R6, #0
MOV    R7, #0
MOV    R8, #0
MOV    R9, #0
MOV    R10, #0
MOV    R11, #0
MOV    R12, #0
MOV    R13, #0
MOV    R14, #0

CPS    #0x11                // Change to FIQ mode.
MOV    R8, #0
MOV    R9, #0
MOV    R10, #0
MOV    R11, #0
MOV    R12, #0
```

```

MOV    R13, #0
MOV    R14, #0

CPS    #0x12                // Change to IRQ mode.
MOV    R13, #0
MOV    R14, #0

CPS    #0x1F                // Change to System mode.
MOV    R13, #0              // System and User modes reuse the same banking
MOV    R14, #0              // of r13 and r14.

CPS    #0x17                // Change to Abort mode.
MOV    R13, #0
MOV    R14, #0

CPS    #0x1B                // Change to Undef mode.
MOV    R13, #0
MOV    R14, #0

CPS    #0x16                // Change to Monitor mode.

MOV    R13, #0
MOV    R14, #0

MOV    R0, #0                // Use MSR in Monitor Mode.
MSR    SP_hyp, R0            // Initialize Hyp mode R13.

```

If a processor implements NEON technology and FP extensions, floating-point registers must be initialized as well.

Example 4-5 shows you how to initialize floating-point registers after reset.

Example 4-5 Floating-point registers initialization

```

// Enable access to FP registers.
MOV    R1, #(0xF << 20)
MCR    P15, 0, R1, C1, C0, 2    // CPACR full access to cp11 and cp10.
MOV    R1, #(0x1 << 30)

// Enable Floating point and Neon unit.
VMSR    FPEXC, R1                // Set FPEXC.EN.

```

ISB // Ensure the enable operation takes effect.

MOV R1, #0
MOV R2, #0
VMOV.F64 D0, R1, R2
VMOV.F64 D1, D0
VMOV.F64 D2, D0
VMOV.F64 D3, D0
VMOV.F64 D4, D0
VMOV.F64 D5, D0
VMOV.F64 D6, D0
VMOV.F64 D7, D0
VMOV.F64 D8, D0
VMOV.F64 D9, D0
VMOV.F64 D10, D0
VMOV.F64 D11, D0
VMOV.F64 D12, D0
VMOV.F64 D13, D0
VMOV.F64 D14, D0
VMOV.F64 D15, D0
VMOV.F64 D16, D0
VMOV.F64 D17, D0
VMOV.F64 D18, D0
VMOV.F64 D19, D0
VMOV.F64 D20, D0
VMOV.F64 D21, D0
VMOV.F64 D22, D0
VMOV.F64 D23, D0
VMOV.F64 D24, D0
VMOV.F64 D25, D0
VMOV.F64 D26, D0
VMOV.F64 D27, D0
VMOV.F64 D28, D0
VMOV.F64 D29, D0
VMOV.F64 D30, D0
VMOV.F64 D31, D0

4.2.2 Initializing stack pointer registers

The stack pointer register (r13) is implicitly used in some instructions, for example, push and pop. You must initialize it with a proper value before using it.

In an MPCore system, different *Stack Pointers* (SPs) must point to different memory addresses to avoid overwriting the stack area. If SPs are used in different modes, you must initialize all of them.

Example 4-6 initializes an SP for one mode. The stack that is pointed to by the SP is located at *stack_top*, and the stack size is *CPU_STACK_SIZE* bytes.

Example 4-6 SP initialization

```
// Initialize the stack pointer.
LDR    R13, =stack_top
ADD    R13, R13, #4
MRC    P15, 0, R0, C0, C0, 5    // Read MPIDR.
AND    R0, R0, #0xFF            // R0 == core number.
MOV    R2, #CPU_STACK_SIZE
MUL    R1, R0, R2                // Create separate stack spaces
SUB    R13, R13, R1              // for each processor.
```

4.2.3 Initializing system control registers

For some system control registers, such as the *Saved Program Status Register* (SPSR) and *Exception Link Register Hype mode* (ELR_hyp), the architecture does not define reset values for them. Therefore, you must initialize the registers before using them.

Example 4-7 shows you how to initialize SPSR and ELR_hyp in Monitor mode.

Example 4-7 SPSR and ELR_hyp initialization

```
// Initialize SPSR in all modes.
MOV    R0, #0
MSR    SPSR, R0
MSR    SPSR_svc, R0
MSR    SPSR_und, R0
MSR    SPSR_hyp, R0
MSR    SPSR_abt, R0
MSR    SPSR_irq, R0
MSR    SPSR_fiq, R0

// Initialize ELR_hyp.
MOV    R0, #0
MSR    ELR_hyp, R0
```

Example 4-7 does not cover all system registers that must be initialized. Theoretically, you must initialize all system registers that do not have architecturally defined reset values.

However, some registers can have IMPLEMENTATION-DEFINED reset values, depending on the implementation of a particular processor. For details, see the section, *General system control registers*, in the *ARM® Architecture Reference Manual ARMv8, for ARMv8-A architecture profile* and the *Technical Reference Manual (TRM)* of the relevant processor.

4.3 Configuring the MMU and caches

The MMU and Cache configuration involves the following operations:

- *Cleaning and invalidating* the caches on page 21.
- *Setting up the MMU* on page 22.
- *Enabling the MMU and caches* on page 27.

4.3.1 Cleaning and invalidating the caches

The content in cache RAM is invalid after reset, so you must perform invalidation operations to initialize all caches in a processor.

In some ARMv7-A processors such as the Cortex-A9 processor, you must use software to invalidate all cache RAMs. In ARMv8-A processors and most ARMv7-A processors, you do not have to do this because hardware automatically invalidates all cache RAMs after reset. However, you must use software to clean and invalidate data cache in some situations, such as the core powerdown process.

Example 4-8 shows you how to clean and invalidate L1 data cache by using looped DCCISW instructions. You can easily modify the code for other level caches or other cache operations.

Example 4-8 Clean and invalidate L1 data cache

```
// Disable L1 Caches.
MRC    P15, 0, R1, C1, C0, 0    // Read SCTLR.
BIC    R1, R1, #(0x1 << 2)      // Disable D Cache.
MCR    P15, 0, R1, C1, C0, 0    // Write SCTLR.

// Invalidate Data cache to create general-purpose code. Calculate the
// cache size first and loop through each set + way.
MOV     R0, #0x0                 // R0 = 0x0 for L1 dcache 0x2 for L2 dcache.
MCR     P15, 2, R0, C0, C0, 0    // CSSELR Cache Size Selection Register.
MRC     P15, 1, R4, C0, C0, 0    // CCSIDR read Cache Size.
AND     R1, R4, #0x7
ADD     R1, R1, #0x4             // R1 = Cache Line Size.
LDR     R3, =0x7FFF
AND     R2, R3, R4, LSR #13      // R2 = Cache Set Number - 1.
LDR     R3, =0x3FFF
AND     R3, R3, R4, LSR #3       // R3 = Cache Associativity Number - 1.
CLZ     R4, R3                  // R4 = way position in CISW instruction.

MOV     R5, #0                  // R5 = way loop counter.
way_loop:
MOV     R6, #0                  // R6 = set loop counter.
set_loop:
ORR     R7, R0, R5, LSL R4       // Set way.
```

ORR	R7, R7, R6, LSL R1	// Set set.
MCR	P15, 0, R7, C7, C6, 2	// DCCISW R7.
ADD	R6, R6, #1	// Increment set counter.
CMP	R6, R2	// Last set reached yet?
BLE	set_loop	// If not, iterate set_loop,
ADD	R5, R5, #1	// else, next way.
CMP	R5, R3	// Last way reached yet?
BLE	way_loop	// if not, iterate way_loop.

4.3.2 Setting up the MMU

ARMv8-A processors use VMSAv8-32 to perform the following operations in AArch32:

- Translate physical address to virtual address.
- Determine memory attributes and check access permission.

Address translation is defined by the translation table and managed by the *Memory Management Unit* (MMU). Before enabling the MMU, you must set up the translation table and translation table walk rules.

Every *Privilege Level* (PL) has dedicated translation tables and control registers. You must set up all translation tables and control registers before use.

For details, see the section, *About VMSAv8-32*, in the *ARM® Architecture Reference Manual ARMv8, for ARMv8-A architecture profile*.

AArch32 supports two translation table formats:

- The VMSAv8-32 short-descriptor format.
- The VMSAv8-32 long-descriptor format.

In ARMv8-A, the hierarchy of software execution privilege, within a Security state, is defined by the *Exception Level* (EL). For relationship between PLs and ELs, please see the section, *Execution privilege, Exception levels, and AArch32 Privilege levels*, in *ARM Architecture Reference Manual ARMv8, for ARMv8-A architecture profile*.

VMSAv8-32 short-descriptor format

The short-descriptor format uses 32-bit descriptor entries in the translation tables, and supports:

- 32-bit input addresses.
- Output addresses of up to 40 bits.
- Address lookup of up to two levels.
- 4KB granule size.

You can use the short-descriptor format only in stage 1 translation at PL0 and PL1. For details, see the section, *The VMSAv8-32 Short-descriptor translation table format*, in the *ARM® Architecture Reference Manual ARMv8, for ARMv8-A architecture profile*.

Example 4-9 uses the short-descriptor format to build a translation table covering 4GB memory space.

- 0-1GB is configured as Normal Cacheable memory.
- 1-4GB is configured as Device-nGnRnE memory.

The translation table contains 4096 x 1MB sections, and is placed at the address defined by TTBR0.

In this translation table, TEX is remapped and the access flag feature is not used.

Example 4-9 Translation table using the VMSAv8-32 short-descriptor format

```
// Initialize TTBCR.
MOV    R0, #0                // Use short descriptor.
MCR     P15, 0, R0, C2, C0, 2 // Base address is 16KB aligned.
                                // Perform translation table walk for TTBR0.

// Initialize DACR.
LDR     R1, =0x55555555      // Set all domains as clients.
MCR     P15, 0, R1, C3, C0, 0 // Accesses are checked against the
                                // permission bits in the translation tables.

// Initialize SCTLR.AFE.
MRC     P15, 0, R1, C1, C0, 0 // Read SCTLR.
BIC     R1, R1, #(0x1 <<29)  // Set AFE to 0 and disable Access Flag.
MCR     P15, 0, R1, C1, C0, 0 // Write SCTLR.

// Initialize TTBR0.
LDR     R0, =ttb0_base       // ttb0_base must be a 16KB-aligned address.
MOV     R1, #0x2B            // The translation table walk is normal, inner
ORR     R1, R0, R1           // and outer cacheable, WB WA, and inner
MCR     P15, 0, R1, C2, C0, 0 // shareable.

// Set up translation table entries in memory
LDR     R4, =0x00100000      // Increase 1MB address each time.
LDR     R2, =0x00015C06      // Set up translation table descriptor with
                                // Secure, global, full accessibility,
                                // executable.
                                // Domain 0, Shareable, Normal cacheable memory
LDR     R3, =1024            // executes the loop 1024 times to set up
                                // 1024 descriptors to cover 0-1GB memory.

Loop:
STR     R2, [R0], #4         // Build a page table section entry.
ADD     R2, R2, R4           // Update address part for next descriptor.
SUBS    R3, #1
BNE     Loop

LDR     R2, =0x40010C02      // Set up translation table descriptors with
                                // secure, global, full accessibility,
                                // Domain=0 Shareable Device-nGnRnE Memory.
LDR     R3, =3072            // Executes loop 3072 times to set up 3072
```

```

// descriptors to cover 1-4GB memory.

loop2:
STR    R2, [R0], #4           // Build a translation table section entry.
ADD    R2, R2, R4             // Update address part for next descriptor.
SUBS   R3, #1
BNE    loop2

```

VMSAv8-32 long-descriptor format

The long-descriptor format uses 64-bit descriptor entries in the translation tables, and supports:

- Input and output addresses of up to 40 bits.
- Address lookup of up to three levels.
- 4KB granule size.

You can use the long-descriptor format for all PLs and stages translation. For details, see the section, *The VMSAv8-32 Long-descriptor translation table format*, in the *ARM® Architecture Reference Manual ARMv8, for ARMv8-A architecture profile*.

Example 4-10 and Example 4-11 use the long-descriptor format to build a translation table covering 4GB memory space:

- 0-1GB memory is configured as Normal Cacheable memory.
- 1-4GB memory is configured as Device-nGnRnE memory.

The translation table contains 512 level2 blocks of 2MB size and 3 level1 blocks of 1GB size.

Example 4-10 initializes translation table control registers, and then uses looped store instructions to build a translation table, which is easier to port.

Example 4-10 Translation table using the VMSAv8-64 long-descriptor format

```

// Initialize translation table control registers

LDR    R1, =0xFF440400        // ATTR0 is Device-nGnRnE. ATTR1 is Device.
                                // ATTR2 is Normal Non-Cacheable.
                                // ATTR3 is Normal Cacheable.

MCR    P15, 0, R1, C10, C2, 0 // Only use MAIR0.

LDR    R0, =0xB0003500        // Use TTBR0 and long descriptor format.
MCR    P15, 0, R0, C2, C0, 2  // translation table walk is Inner-shareable
                                // Normal Inner and Outer cacheable.

LDR    R0, =ttb0_base
MOV    R1, #0
MCRR   P15, 0, R0, R1, C2     // TTBR0 ASID=0.

// Set up translation table entries in memory with looped store instructions.
// Set a level 1 translation table.

```

```

// The first entry points to level2_pagetable.
LDR    R1, =level2_pagetable // Must be a 4KB-aligned address.
LDR    R2, =0xFFFFF000
AND    R2, R1, R2
ORR    R2, R2, #0x3
MOV    R3, #0                // NSTable=0 APTable=0 XNTTable=0 PXNTable=0.
STRD   R2, R3, [R0], #8
// The second entry is 1GB block, 0x40000000 - 0x7FFFFFFF.
MOV    R3, #0                // XN=0 PXN=0.
LDR    R2, =0x40000741       // nG=0 AF=1 Inner and Outer Shareable.
STRD   R2, R3, [R0], #8     // R/W at all ELs secure memory.
// The third entry is 1GB block, 0x80000000 - 0xBFFFFFFF.
LDR    R2, =0x80000741       // AttrIdx=000 Device-nGnRnE.
STRD   R2, R3, [R0], #8
// The fourth entry is 1GB block, 0xC0000000 - 0xFFFFFFFF.
LDR    R2, =0xC0000741       // AttrIdx=000 Device-nGnRnE.
STRD   R2, R3, [R0], #8

// Set level 2 translation table.
LDR    R0, =level2_pagetable // R0 is the base address of level2_pagetable.
LDR    R2, =0x0000074D       // nG=0 AF=1 Inner and Outer Shareable.
                                     // R/W at all ELs secure memory.
                                     // AttrIdx=011 Normal Cacheable.
MOV    R3, #0                // XN=0 PXN=0.
MOV    R4, #512              // Set 512 level2 block entries.
LDR    R5, =0x00200000       // Increase 2MB address each time.
loop:
STRD   R2, R3, [R0], #8     // Each entry occupies two words.
ADD    R2, R2, R5
SUBS   R4, #1
BNE    loop

```

Example 4-11 creates a section as a translation table at compile time. This method is fast for simulations. It is written with the GNU assembly grammar. The code to initialize translation table control registers in example 4-10 is still required.

```
// Put a 64-bit value with little endianness.
.macro PUT_64B high, low
.word \low
.word \high
.endm

// Create an entry pointing to a next-level table.
.macro TABLE_ENTRY PA, ATTR
PUT_64B \ATTR, ((\PA) + 0x3)
.endm

// Create an entry for a 1GB block.
.macro BLOCK_1GB PA, ATTR_HI, ATTR_LO
PUT_64B \ATTR_HI, ((\PA) & 0xC0000000) | \ATTR_LO | 0x1
.endm

// Create an entry for a 2MB block.
.macro BLOCK_2MB PA, ATTR_HI, ATTR_LO
PUT_64B \ATTR_HI, ((\PA) & 0xFFE00000) | \ATTR_LO | 0x1
.endm

.align 12
ttb0_base:
TABLE_ENTRY level2_pagetable, 0
BLOCK_1GB 0x40000000, 0, 0x740
BLOCK_1GB 0x80000000, 0, 0x740
BLOCK_1GB 0xC0000000, 0, 0x740

.align 12
level2_pagetable:
.set ADDR, 0x000 // The current page address.
.rept 0x200
BLOCK_2MB (ADDR << 20), 0, 0x74C
.set ADDR, ADDR+2
.endr
```

4.3.3 Enabling the MMU and caches

You must initialize the MMU and caches before enabling them. You must set the SMPEN bit before enabling the MMU and cache for all ARMv8-A processors, to support hardware coherency.

Example 4-12 shows you how to set the SMPEN bit and enable the MMU and caches.

Example 4-12 SMPEN bit setting and the MMU and cache enablement

```
// SMP is implemented in the CPUECTLR register.
MRRC    P15, 1, R0, R1, C15          // Read CPUECTLR.
ORR     R0, R0, #(0x1 << 6)         // Set SMPEN.
MCRR    P15, 1, R0, R1, C15          // Write CPUECTLR.

// Enable caches and the MMU.
MRC     P15, 0, R1, C1, C0, 0        // Read SCTLR.
ORR     R1, R1, #(0x1 << 2)          // The C bit (data cache).
ORR     R1, R1, #(0x1 << 12)         // The I bit (instruction cache).
ORR     R1, R1, #0x1                 // The M bit (MMU).
MCR     P15, 0, R1, C1, C0, 0        // Write SCTLR.
DSB
ISB
```

4.4 Enabling NEON and Floating Point

In AArch32 mode, access to NEON technology and FP functionality is disabled by default, so it must be explicitly enabled. For details, see the section, *Enabling Advanced SIMD and floating-point support*, in the *ARM® Architecture Reference Manual ARMv8, for ARMv8-A architecture profile*.

This section describes how to enable general NEON technology and FP functionality in both the Secure world and the Non-secure world.

4.4.1 Enabling general NEON and FP functionality

Example 4-13 shows you how to enable general NEON technology and FP functionality after reset.

Example 4-13 NEON and FP function enablement

```
// Enable access to NEON/FP by enabling access to Coprocessors 10 and 11.
// Enable Full Access in both privileged and non-privileged modes.
MOV    R0, #(0xF << 20)      // Enable CP10 & CP11 function
MCR    P15, 0, R0, C1, C0, 2 // Write the Coprocessor Access Control
ISB                               // Register (CPACR).

// Switch on the FP and NEON hardware.
MOV    R1, #(0x1 << 30)
VMSR   FPEXC, R1
```

4.4.2 Enabling access to the NEON and FP functionality in the Non-secure world

Access to NEON technology and FP functionality from the Non-secure world is disabled after reset. If software requires access to the NEON and FP registers in the Non-secure world, *Non-secure Access Control Register* (NSACR) must be initialized in EL3.

Example 4-14 shows you how to configure the NSACR after reset.

Example 4-14 NSACR configuration

```
// Enable access NEON/FP in Non-secure world.
MOV    R1, #(0x3 << 10)      // Enable Non-secure access to CP10 & CP11.
MCR    P15, 0, R1, C1, C1, 2 // Write NSACR.
```

4.4.3 Enabling access to the NEON and FP functionality in Non-secure EL1 and EL0

Access to the NEON and FP functionality from Non-secure EL1 or EL0 can be trapped to Hypervisor mode. The trap must be disabled if a program must access NEON and FP functionality in Non-secure EL1 or EL0. The trap function is disabled by default after core reset, so this step might be unnecessary.

Example 4-15 shows you how to disable trap of accesses to NEON technology and FP functionality from Non-secure EL1 or EL0 by programming the *Hyp Architectural Feature Trap Register* (HCPTR) register.

Example 4-15 Enable access to the NEON and FP function

```
// Enable access to NEON and FP in Non-secure EL1 and EL0.  
LDR    R1, =0x33FF  
MCR    P15, 4, R1, C1, C1, 2 // Write HCPTR.
```

Note

The HCPTR register can be accessed in EL2 and EL3 (NS=1).

4.5 Changing modes

If the Security Extension is implemented, AArch32 has two security states and nine processor modes:

- Security states:
 - Secure state.
 - Non-secure state.
- Processor modes
 - User.
 - System.
 - FIQ.
 - IRQ.
 - Supervisor.
 - Abort.
 - Undefined.
 - Hyp
 - Monitor.

The following figure shows how the security states and processor modes are structured and their relationship with Exception levels in AArch32.

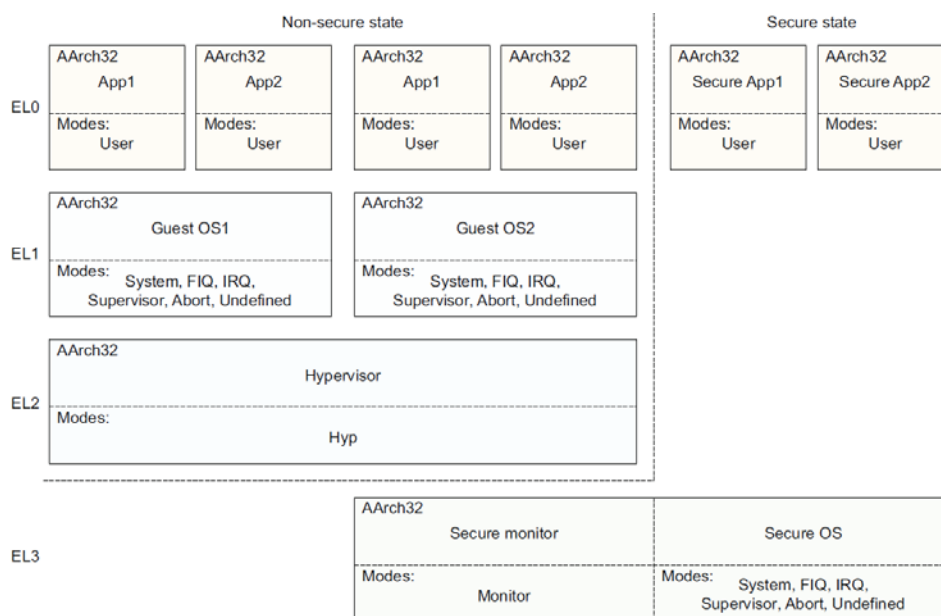


Figure 4-1 Security states and processor modes

For details, see the section, *Security state*, in the *ARM® Architecture Reference Manual ARMv8, for ARMv8-A architecture profile*.

The following sections describe how to change between these modes when a processor runs in AArch32:

- *Changing between User, System, FIQ, IRQ, Supervisor, Abort, Undefined modes* on page 31.
- *Changing between Secure world and Non-secure world* on page 31.

- *Changing between Hypervisor mode and other modes* on page 33.

4.5.1 Changing between User, System, FIQ, IRQ, Supervisor, Abort, Undefined modes

When booting in AArch32 mode, processors enter secure Supervisor mode after reset.

Normally, processors take or return exceptions to change to other modes. To simplify the test, it can be done by directly changing the CPSR.M bits in a bare-metal test.

Example 4-16 shows you how to change from a non-User mode to other modes.

Example 4-16 Mode change

```
.equ    Mode_USR,    0x10
.equ    Mode_FIQ,    0x11
.equ    Mode_IRQ,    0x12
.equ    Mode_SVC,    0x13
.equ    Mode_MNT,    0x16
.equ    Mode_ABT,    0x17
.equ    Mode_HYP,    0x1A
.equ    Mode_UND,    0x1B
.equ    Mode_SYS,    0x1F

// When a processor is in Monitor, System, FIQ, IRQ, Supervisor, Abort
// or Undefined mode, use the CPS instruction to change modes.
CPS     #Mode_FIQ
```

Example 4-17 shows you how to change from User mode to Supervisor mode.

Example 4-17 Mode switch from User mode to Supervisor mode

```
// When processors are in User mode, use SVC to change from User mode
// to SVC mode. Make sure that VBAR is initialized before executing SVC.
SVC     #0
```

4.5.2 Changing between the Secure world and Non-secure world

All transitions between Secure and Non-secure world pass through Monitor mode. Therefore, to change Security status, you must first execute an SMC instruction to enter Monitor mode.

————— Note —————

Monitor mode belongs in the Secure world.

Example 4-18 shows you how to use the SMC instruction to enter Monitor mode.

Example 4-18 Enter Monitor mode

```
// Use an SMC to change to Monitor mode.
// Make sure that MVBAR is initialized before executing the SMC.
SMC    #0
```

To switch from the Secure world to the Non-secure world, the processor must set SCR.NS to 1 in Monitor mode. After that, the processor returns to Non-secure world with an exception return.

Example 4-19 shows you how to switch to Non-secure Supervisor mode when the processor is in Monitor mode.

Example 4-19 Switch from Secure world to Non-secure world

```
// Use an exception return in the Monitor exception handler to
// enter the Non-secure world.
MRC    P15, 0, R1, C1, C1, 0    // Read Secure Configuration Register
                                      // (SCR).
ORR     R1, R1, #(1 << 0)       // Set SCR.NS (bit 0).
BIC     R1, R1, #(1 << 7)       // Clear SCR.SCD (bit 7).
MCR     P15, 0, R1, C1, C1, 0    // Write SCR.

// Initialize registers to save values.
MOV     R0, #0
MCR     P15, 0, R0, C1, C0, 0    // SCTLR(NS).
LDR     R1, =vector_table_base_address
MCR     P15, 0, R1, C12, C0, 0    // VBAR(NS).
// Exception return.
MSR     SPSR_cxsf, #Mode_SVC     // entering supervisor mode(NS).
LDR     R14, =SVC_entry          // SVC_entry points to the first
                                      // instruction of SVC mode code.

ERET
```

To switch from the Non-secure world to the Secure world, the processor performs the following steps:

1. Enter Monitor mode.
2. Set SCR.NS to 0 in Monitor mode.
3. Switch to other modes in the Secure world.

Example 4-20 shows you how to clear the SCR.NS bit when the processor is in Monitor mode.

```

MRC    P15, 0, R1, C1, C1, 0    // Read SCR.
BIC    R1, R1, #(1 << 0)        // Set SCR.NS (bit 0).
MCR    P15, 0, R1, C1, C1, 0    // Write SCR.

```

4.5.3 Changing between Hypervisor mode and other modes

To enter Hypervisor mode, use an exception return from Monitor mode (NS=1) or take an exception in any of the Non-secure System, FIQ, IRQ, Supervisor, Abort, or Undefined modes.

Example 4-21 shows you how to enter Hypervisor mode from Monitor mode.

Example 4-21 Switch from Monitor mode to Hypervisor mode

```

// Enter Hypervisor mode by using an exception return when the processor
// is in Monitor mode.
MRC    P15, 0, R1, C1, C1, 0    // Read SCR.
ORR    R1, R1, #(1 << 0)        // Set SCR.NS (bit 0).
ORR    R1, R1, #(1 << 8)        // Set SCR.HCE (bit 8) and enable HVC.
MCR    P15, 0, R1, C1, C1, 0    // Write SCR.

// Initialize registers to save values before changing to Hypervisor mode.
MOV    R0, #0
MCR    P15, 4, R0, C1, C0, 0    // HSCTLR.
MCR    P15, 4, R0, C1, C1, 0    // HCR.
MCR    P15, 4, R0, C1, C1, 4    // HCR2.
LDR    R1, =hyp_vector_table_base_address
MCR    P15, 4, R1, C12, C0, 0    // HVBAR.

MSR    SPSR_cxsf, #Mode_HYP
LDR    R14, =Hyp_entry          // Hyp_entry points to the first
                                // instruction of Hypervisor mode code.

ERET

```

Example 4-22 shows you how to enter Hypervisor mode from any of the Non-secure System, FIQ, IRQ, Supervisor, Abort, or Undefined modes.

Example 4-22 Enter Hypervisor mode

```

// Use an HVC to call hypervisor exception.
// Make sure that HVBAR is initialized before executing the HVC.
HVC    #0

```

To exit Hypervisor mode, use an SMC instruction to enter Monitor mode or use an exception to return to Non-secure EL1 or EL0 mode, see *Changing between the Secure world and Non-secure world*.

5 Boot code for AArch64 mode

Read this chapter for boot code examples for AArch64.

It contains the following topics:

- *Initializing exceptions* on page 36.
- *Initializing registers* on page 41.
- *Configuring the MMU and caches* on page 45.
- *Enabling NEON and Floating Point* on page 50.
- *Changing Exception levels* on page 51.

5.1 Initializing exceptions

Exception initialization requires:

- Setting up the vector table.
- Asynchronous exceptions routing and masking configurations.

5.1.1 Setting up a vector table

In AArch64, a reset vector is no longer part of the exception vector table. There are dedicated configure input pins and registers for the reset vector. Other exception vectors are stored in the vector table.

Reset vector

In AArch64, the processor starts execution from an IMPLEMENTATION-DEFINED address, which is defined by the hardware input pins **RVBARADDR** and can be read by the RVBAR_EL3 register. You must place boot code at this address.

Vector table

There are dedicated vector tables for each exception level:

- VBAR_EL3.
- VBAR_EL2.
- VBAR_EL1.

The vector table in AArch64 is different from that in AArch32. The vector table in AArch64 mode contains 16 entries. Each entry is 128B in size and contains at most 32 instructions. Vector tables must be placed at a 2KB-aligned address. The addresses are specified by initializing VBAR_ELn registers.

For more details about the vector table, see the section, *Exception vectors*, in the *ARM® Architecture Reference Manual ARMv8, for ARMv8-A architecture profile*.

The following figure shows you how the vector table is structured.

0x780	SError/vSError	Lower EL using AArch32
0x700	FIQ/vFIQ	
0x680	IRQ/vIRQ	
0x600	Synchronous	
0x580	SError/vSError	Lower EL using AArch64
0x500	FIQ/vFIQ	
0x480	IRQ/vIRQ	
0x400	Synchronous	
0x380	SError/vSError	Current EL with SPx
0x300	FIQ/vFIQ	
0x280	IRQ/vIRQ	
0x200	Synchronous	
0x180	SError/vSError	Current EL with SP0
0x100	FIQ/vFIQ	
0x080	IRQ/vIRQ	
VBAR_ELn + 0x000	Synchronous	

Figure 5-1 vector table structure

Example 5-1 shows you how to initialize VBAR_EL3, VBAR_EL2, and VBAR_EL1 after reset.

Example 5-1 Vector Base Address registers initialization

```
// Initialize VBAR_EL3.
LDR    X1, = vector_table_el3
MSR    VBAR_EL3, X1

LDR    X1, = vector_table_el2
MSR    VBAR_EL2, X1

LDR    X1, = vector_table_el1
MSR    VBAR_EL1, X1
```

Example 5-2 shows a typical vector table for exceptions in AArch64.

Example 5-2 Exception vector table

```
// Typical exception vector table code.
.balign 0x800
Vector_table_el3:
curr_el_sp0_sync:    // The exception handler for the synchronous
                    // exception from the current EL using SP0.

.balign 0x80
curr_el_sp0_irq:     // The exception handler for the IRQ exception
                    // from the current EL using SP0.

.balign 0x80
curr_el_sp0_fiq:     // The exception handler for the FIQ exception
                    // from the current EL using SP0.

.balign 0x80
curr_el_sp0_serror:  // The exception handler for the system error
                    // exception from the current EL using SP0.

.balign 0x80
curr_el_spx_sync:    // The exception handler for the synchronous
                    // exception from the current EL using the
                    // current SP.

.balign 0x80
curr_el_spx_irq:     // The exception handler for IRQ exception
                    // from the current EL using the current SP.
```

```

.balign 0x80
curr_el_spx_fiq:      // The exception handler for the FIQ exception
                      // from the current EL using the current SP.

.balign 0x80
curr_el_spx_serror:   // The exception handler for the system error
                      // exception from the current EL using the
                      // current SP.

.balign 0x80
lower_el_aarch64_sync: // The exception handler for the synchronous
                      // exception from a lower EL (AArch64).

.balign 0x80
lower_el_aarch64_irq: // The exception handler for the IRQ exception
                      // from a lower EL (AArch64).

.balign 0x80
lower_el_aarch64_fiq: // The exception handler for the FIQ exception
                      // from a lower EL (AArch64).

.balign 0x80
lower_el_aarch64_serror: // The exception handler for the system error
                        // exception from a lower EL(AArch64).

.balign 0x80
lower_el_aarch32_sync: // The exception handler for the synchronous
                      // exception from a lower EL(AArch32).

.balign 0x80
lower_el_aarch32_irq:  // The exception handler for the IRQ exception
                      // from a lower EL (AArch32).

.balign 0x80
lower_el_aarch32_fiq:  // The exception handler for the FIQ exception
                      // from a lower EL (AArch32).

.balign 0x80
lower_el_aarch32_serror: // The exception handler for the system error
                        // exception from a lower EL(AArch32).

```

5.1.2 Enabling asynchronous exceptions

Asynchronous exceptions including SError, IRQ and FIQ. They are default masked after reset. Therefore, if SError, IRQ and FIQ are to be taken, the routing rules must be set and the mask must be cleared.

To enable interrupts, you must also initialize the external interrupt to deliver the interrupt to the processor, but it is not covered in this document.

Asynchronous exceptions routing

Asynchronous exception routing determines which Exception level is used to handle an asynchronous exception.

To route an asynchronous exception to EL3, you must set SCR_EL3.{EA,IRQ,FIQ}.

Example 5-3 shows how to route SError, IRQ and FIQ to EL3.

Example 5-3 SError, IRQ and FIQ routing enablement in EL3

```
MRS    X0, SCR_EL3
ORR     X0, X0, #(1<<3)    // The EA bit.
ORR     X0, X0, #(1<<1)    // The IRQ bit.
ORR     X0, X0, #(1<<2)    // The FIQ bit.
MSR     SCR_EL3, X0
```

To route an asynchronous exception to EL2 rather than EL3, you must set HCR_EL2.{AMO,FMO,IMO} and clear SCR_EL3.{EA,IRQ,FIQ}.

Example 5-4 shows you how to route SError, IRQ and FIQ to EL2.

Example 5-4 SError, IRQ and FIQ routing enablement in EL2

```
MRS     X0, HCR_EL2
ORR      X0, X0, #(1<<5)    // The AMO bit.
ORR      X0, X0, #(1<<4)    // The IMO bit.
ORR      X0, X0, #(1<<3)    // The FMO bit.
MSR      HCR_EL2, X0
```

If an interrupt is not routed to EL3 or EL2, it is routed to EL1 by default.

Asynchronous exceptions mask

Whether an asynchronous exception is masked depends on the following factors:

- The target Exception level to which the interrupt is routed.
- The PSTATE.{A,I,F} value.

When a target Exception level is lower than the current Exception level, the asynchronous exception is masked implicitly, regardless of the PSTATE.{A,I,F} value.

When a target Exception level is same as the current Exception level, the asynchronous exception is masked if PSTATE.{A,I,F} is 1.

When a target Exception level is higher than the current Exception level and the target Exception level is EL2 or EL3, the asynchronous exception is taken, regardless of the PSTATE.{A,I,F} value.

When a target Exception level is higher than the current Exception level and the target Exception level is EL1, the asynchronous exception is masked if PSTATE.{A,I,F} is 1.

Example 5-5 shows you how to clear the mask of SError, IRQ and FIQ in PSTATE.

Example 5-5 Enable SError, IRQ and FIQ

```
// Enable SError, IRQ and FIQ
MSR    DAIFClr, #0x7
```

For more details about enabling asynchronous exceptions, see the section, *Asynchronous exception types, routing, masking and priorities*, in the *ARM® Architecture Reference Manual ARMv8, for ARMv8-A architecture profile*.

5.2 Initializing registers

Register initialization involves initializing the following registers:

- General-purpose registers.
- Stack pointer registers.
- System control registers.

5.2.1 Initializing general purpose registers

ARM processors use some non-reset flip-flops. This can cause X-propagation issues in simulations. Register initialization helps reduce the possibility of the issue.

————— Note —————

This initialization is not required on silicon chips because X status only exists in hardware simulations.

Example 5-6 shows you how to initialize general-purpose registers after reset.

Example 5-6 Register bank initialization

```
// Initialize the register bank.
MOV    X0,  XZR
MOV    X1,  XZR
MOV    X2,  XZR
MOV    X3,  XZR
MOV    X4,  XZR
MOV    X5,  XZR
MOV    X6,  XZR
MOV    X7,  XZR
MOV    X8,  XZR
MOV    X9,  XZR
MOV    X10, XZR
MOV    X11, XZR
MOV    X12, XZR
MOV    X13, XZR
MOV    X14, XZR
MOV    X15, XZR
MOV    X16, XZR
MOV    X17, XZR
MOV    X18, XZR
MOV    X19, XZR
MOV    X20, XZR
MOV    X21, XZR
MOV    X22, XZR
```

MOV	X23, XZR
MOV	X24, XZR
MOV	X25, XZR
MOV	X26, XZR
MOV	X27, XZR
MOV	X28, XZR
MOV	X29, XZR
MOV	X30, XZR

If a processor implements the NEON and FP extension, floating-point registers must be initialized as well.

Example 5-7 shows you how to initialize floating-point registers after reset.

Example 5-7 Floating-point registers initialization

MSR	CPTR_EL3, XZR
MSR	CPTR_EL2, XZR

FMOV	D0, XZR
FMOV	D1, XZR
FMOV	D2, XZR
FMOV	D3, XZR
FMOV	D4, XZR
FMOV	D5, XZR
FMOV	D6, XZR
FMOV	D7, XZR
FMOV	D8, XZR
FMOV	D9, XZR
FMOV	D10, XZR
FMOV	D11, XZR
FMOV	D12, XZR
FMOV	D13, XZR
FMOV	D14, XZR
FMOV	D15, XZR
FMOV	D16, XZR
FMOV	D17, XZR
FMOV	D18, XZR
FMOV	D19, XZR
FMOV	D20, XZR

```

FMOV    D21, XZR
FMOV    D22, XZR
FMOV    D23, XZR
FMOV    D24, XZR
FMOV    D25, XZR
FMOV    D26, XZR
FMOV    D27, XZR
FMOV    D28, XZR
FMOV    D29, XZR
FMOV    D30, XZR
FMOV    D31, XZR

```

5.2.2 Initializing stack pointer registers

The stack pointer register is implicitly used in some instructions, for example, push and pop. You must initialize it with a proper value before using it.

In an MPCore system, different stack pointers must point to different memory addresses to avoid overwriting the stack area. If SPs in different Exception levels are used, you must initialize all of them.

Example 5-8 shows you how to initialize an SP for the current Exception level. The stack pointed to by the SP is at *stack_top*, and the stack size is *CPU_STACK_SIZE* bytes.

Example 5-8 SP initialization in the current Exception level

```

// Initialize the stack pointer.
ADR    X1, stack_top
ADD    X1, X1, #4
MRS    X2, MPIDR_EL1
AND    X2, X2, #0xFF           // X2 == CPU number.
MOV    X3, #CPU_STACK_SIZE
MUL    X3, X2, X3              // Create separated stack spaces
SUB    X1, X1, X3              // for each processor
MOV    SP, X1

```

5.2.3 Initializing system control registers

Some system control registers do not have architectural reset values. Therefore, you must initialize the registers based on your software requirements before using them.

Example 5-9 shows how to initialize HCR_EL2, SCTLR_EL2, and SCTLR_EL1 after reset.

Example 5-9 System control registers initialization

```

MSR    HCR_EL2, XZR

```

LDR	X1, =0x30C50838
MSR	SCTLR_EL2, X1
MSR	SCTLR_EL1, X1

This example does not cover all system registers that need initialization. Theoretically, you must initialize all system registers that do not have architecturally defined reset values. However, some registers can have IMPLEMENTATION-DEFINED reset values, depending on the implementation of a particular processor. For details, see the section, *General system control registers*, in the *ARM® Architecture Reference Manual ARMv8, for ARMv8-A architecture profile* and the *TRM* of the relevant processor.

5.3 Configuring the MMU and caches

The MMU and cache configuration involves the following operations:

- *Cleaning and invalidating caches* on page 45.
- *Setting up the MMU* on page 46.
- *Enabling the MMU and caches* on page 49.

5.3.1 Cleaning and invalidating the caches

The content in cache RAM is invalid after reset. ARMv8-A processors implement hardware that automatically invalidates all cache RAMs after reset, so software invalidation is unnecessary after reset. However, cleaning and invalidating data cache is still necessary in some situations, such as the core powerdown process.

Example 5-10 shows you how to clean and invalidate the L1 data cache by using looped DC CISC instructions in EL3. You can easily modify the code for other level caches or other cache operations.

Example 5-10 Clean and invalidate L1 data cache

```
// Disable L1 Caches
MRS    X0, SCTLR_EL3           // Read SCTLR_EL3.
BIC    X0, X0, #(0x1 << 2)    // Disable D Cache.
MSR    SCTLR_EL3, X0           // Write SCTLR_EL3.

// Invalidate Data cache to make the code general purpose.
// Calculate the cache size first and loop through each set +
// way.
MOV     X0, #0x0                // X0 = Cache level
MSR     CSSELR_EL1, x0          // 0x0 for L1 Dcache 0x2 for L2 Dcache.

MRS     X4, CCSIDR_EL1          // Read Cache Size ID.
AND     X1, X4, #0x7
ADD     X1, X1, #0x4            // X1 = Cache Line Size.
LDR     X3, =0x7FFF
AND     X2, X3, X4, LSR #13     // X2 = Cache Set Number - 1.
LDR     X3, =0x3FF
AND     X3, X3, X4, LSR #3      // X3 = Cache Associativity Number - 1.
CLZ     W4, W3                  // X4 = way position in the CISC instruction.

MOV     X5, #0                  // X5 = way counter way_loop.
way_loop:
MOV     X6, #0                  // X6 = set counter set_loop.
set_loop:
LSL     X7, X5, X4
```

```

ORR    X7, X0, X7           // Set way.
LSL    X8, X6, X1
ORR    X7, X7, X8           // Set set.
DC     c1sw, X7             // Clean and Invalidate cache line.
ADD    X6, X6, #1           // Increment set counter.
CMP    X6, X2               // Last set reached yet?
BLE    set_loop             // If not, iterate set_loop,
ADD    X5, X5, #1           // else, next way.
CMP    X5, X3               // Last way reached yet?
BLE    way_loop             // If not, iterate way_loop.

```

5.3.2 Setting up the MMU

ARMv8-A processors use VMSAv8-64 to perform the following operations at AArch64:

- Translate physical address to virtual address.
- Determine memory attributes and check access permission.

Address translation is defined by a translation table and managed by the MMU. Each Exception level has a dedicated translation page table. The translation tables must be set up before enabling the MMU.

VMSAv8-64 uses 64-bit descriptor format entries in the translation tables. It supports

- Up to 48-bit input and output addresses.
- Three granule sizes: 4KB, 16KB, and 64KB.
- Address lookup of up to four levels.

For details, see the section, *The AArch64 Virtual Memory System Architecture*, in the *ARM® Architecture Reference Manual ARMv8, for ARMv8-A architecture profile*.

Example 5-11 and Example 5-12 build an EL3 translation table with a 4KB granule size covering 4GB memory space:

- 0-1GB memory is configured as Normal cacheable memory.
- 1-4GB memory is configured as Device-nGnRnE memory.

The translation table contains 512 level2 blocks of 2MB size and 3 level1 blocks of 1GB size.

Example 5-11 first initializes translation table control registers, and then uses looped store instructions to build a translation table, which is easier to port.

Example 5-11 Build translation tables using looped store instructions

```

// Initialize translation table control registers
LDR    X1, =0x3520          // 4GB space 4KB granularity
                                // Inner-shareable.
MSR    TCR_EL3, X1          // Normal Inner and Outer Cacheable.

LDR    X1, =0xFF440400      // ATTR0 Device-nGnRnE ATTR1 Device.
MSR    MAIR_EL3, X1         // ATTR2 Normal Non-Cacheable.

```

```

// ATTR3 Normal Cacheable.

ADR    X0, ttb0_base           // ttb0_base must be a 4KB-aligned address.
MSR     TTBR0_EL3, X0

// Set up translation table entries in memory with looped store
// instructions.
// Set the level 1 translation table.
// The first entry points to level2_pagetable.
LDR     X1, = level2_pagetable // Must be a 4KB align address.
LDR     X2, =0xFFFFF000
AND     X2, X1, X2             // NSTable=0 APTable=0 XNTable=0 PXNTable=0.
ORR     X2, X2, 0x3
STR     X2, [X0], #8
// The second entry is 1GB block from 0x40000000 to 0x7FFFFFFF.
LDR     X2, =0x40000741        // Executable Inner and Outer Shareable.
STR     X2, [X0], #8           // R/W at all ELs secure memory
// AttrIdx=000 Device-nGnRnE.

// The third entry is 1GB block from 0x80000000 to 0xBFFFFFFF.
LDR     X2, =0x80000741
STR     X2, [X0], #8
// The fourth entry is 1GB block from 0xC0000000 to 0xFFFFFFFF.
LDR     X2, =0xC0000741
STR     X2, [X0], #8

// Set level 2 translation table.
LDR     X0, =level2_pagetable // Base address of level2_pagetable.
LDR     X2, =0x0000074D        // Executable Inner and Outer Shareable.
// R/W at all ELs secure memory.
// AttrIdx=011 Normal Cacheable.
MOV     X4, #512               // Set 512 level2 block entries.
LDR     X5, =0x00200000        // Increase 2MB address each time.
loop:
STR     X2, [X0], #8           // Each entry occupies 2 words.
ADD     X2, X2, X5
SUBS    X4, X4, #1
BNE     loop

```

Example 5-12 creates a section as a translation table at compile time. This method is fast for simulations. It is written with the GNU assembly grammar. The code to initialize translation table control registers in example 5-11 is still required.

Example 5-12 Build translation tables using sections at compile time

```
// Put a 64-bit value with little endianness.
.macro PUT_64B high, low
.word \low
.word \high
.endm

// Create an entry pointing to a next-level table.
.macro TABLE_ENTRY PA, ATTR
PUT_64B \ATTR, ((\PA) + 0x3)
.endm

// Create an entry for a 1GB block.
.macro BLOCK_1GB PA, ATTR_HI, ATTR_LO
PUT_64B \ATTR_HI, ((\PA) & 0xC0000000) | \ATTR_LO | 0x1
.endm

// Create an entry for a 2MB block.
.macro BLOCK_2MB PA, ATTR_HI, ATTR_LO
PUT_64B \ATTR_HI, ((\PA) & 0xFFE00000) | \ATTR_LO | 0x1
.endm

.align 12 // 12 for 4KB granule.
ttb0_base:
TABLE_ENTRY level2_pagetable, 0
BLOCK_1GB 0x40000000, 0, 0x740
BLOCK_1GB 0x80000000, 0, 0x740
BLOCK_1GB 0xC0000000, 0, 0x740

.align 12 // 12 for 4KB granule.
level2_pagetable:
.set ADDR, 0x000 // The current page address.
.rept 0x200
BLOCK_2MB (ADDR << 20), 0, 0x74C
.set ADDR, ADDR+2
```

```
.endr
```

5.3.3 Enabling the MMU and caches

You must initialize the MMU and caches before enabling them. All ARMv8-A processors require the SMPEN bit to be set before enabling the MMU and cache to support hardware coherency.

Example 5-13 shows you how to set the SMPEN bit and enable the MMU and cache.

Example 5-13 Set the SMPEN bit and enable the MMU and Cache

```
// It is implemented in the CPUECTLR register.
MRS    X0, S3_1_C15_C2_1
ORR     X0, X0, #(0x1 << 6)           // The SMP bit.
MSR     S3_1_C15_C2_1, X0

// Enable caches and the MMU.
MRS     X0, SCTL3_EL3
ORR     X0, X0, #(0x1 << 2)           // The C bit (data cache).
ORR     X0, X0, #(0x1 << 12)          // The I bit (instruction cache).
ORR     X0, X0, #0x1                  // The M bit (MMU).
MSR     SCTL3_EL3, X0
DSB     SY
ISB
```

5.4 Enabling NEON and Floating Point

In AArch64, you do not need to enable access to the NEON and FP registers. However, access to the NEON and FP registers can still be trapped.

Example 5-14 shows how to disable access trapping to NEON and FP registers in all Exception levels.

Example 5-14 disable access trapping to NEON and FP registers

```
// Disable trapping of accessing in EL3 and EL2.
MSR    CPTR_EL3, XZR
MSR    CPTR_EL3, XZR

// Disable access trapping in EL1 and EL0.
MOV    X1, #(0x3 << 20)    // FPEN disables trapping to EL1.
MSR    CPACR_EL1, X1
ISB
```

5.5 Changing Exception levels

The ARMv8-A architecture introduces four *Exception levels*.

- EL0.
- EL1.
- EL2.
- EL3.

Sometimes, you must change between these Exception levels in test cases. Processors change Exception levels when an exception is taken or returned. For details about Exception Levels, see the section, *Exception levels*, in the *ARM® Architecture Reference Manual ARMv8, for ARMv8-A architecture profile*.

5.5.1 AArch64 EL3 to AArch64 EL0

Processors enter EL3 after reset. The control register and exception status of lower Exception levels are not defined. To enter a lower Exception level, you must initialize Execution state and control registers, and then use a fake exception return by executing ERET instruction.

Example 5-15 shows how to switch from EL3 to Non-secure EL0.

Example 5-15 Switch from EL3 to Non-secure EL0

```
// Initialize SCTLR_EL2 and HCR_EL2 to save values before entering EL2.
MSR    SCTLR_EL2, XZR
MSR    HCR_EL2, XZR

// Determine the EL2 Execution state.
MRS    X0, SCR_EL3
ORR     X0, X0, #(1<<10) // RW  EL2 Execution state is AArch64.
ORR     X0, X0, #(1<<0)  // NS   EL1 is Non-secure world.
MSR     SCR_EL3, x0
MOV     X0, #0b01001      // DAIF=0000
MSR     SPSR_EL3, X0      // M[4:0]=01001 EL2h must match SCR_EL3.RW

// Determine EL2 entry.
ADR     X0, e12_entry      // e12_entry points to the first instruction of
MSR     ELR_EL3, X0        // EL2 code.

ERET

e12_entry:
// Initialize the SCTLR_EL1 register before entering EL1.
MSR     SCTLR_EL1, XZR

// Determine the EL1 Execution state.
```

```

MRS    X0, HCR_EL2
ORR     X0, X0, #(1<<31)  // RW=1  EL1 Execution state is AArch64.
MSR     HCR_EL2, X0
MOV     X0, #0b00101      // DAIF=0000
MSR     SPSR_EL2, X0      // M[4:0]=00101 EL1h must match HCR_EL2.RW.

ADR     X0, e11_entry     // e11_entry points to the first instruction of
MSR     ELR_EL2, X0       // EL1 code.

ERET

e11_entry:
// Determine the EL0 Execution state.
MOV     X0, #0b00000      // DAIF=0000  M[4:0]=00000  EL0t.
MSR     SPSR_EL1, X0

ADR     x0, e10_entry     // e11_entry points to the first instruction of
MSR     ELR_EL1, X0       // EL0 code.

ERET

e10_entry:
// EL0 code here.

```

5.5.2 AArch64 EL2 to AArch32 EL1

It is possible to have a mix of Execution states in different Exception levels. When a higher Exception level uses AArch64, lower Exception levels are allowed to use either AArch64 or AArch32. Therefore, it is possible to change from higher Exception levels in AArch64 to lower Exception levels in AArch32.

Example 5-16 shows you how to change from AArch64 EL2 to AArch32 EL1.

Example 5-16 Switch from AArch64 EL2 to AArch32 EL1

```

// Initialize the SCTLR_EL1 register before entering EL1.
MSR     SCTLR_EL1, XZR

MRS     X0, HCR_EL2
BIC     X0, X0, #(1<<31)  // RW=0 EL1 Execution state is AArch32.
MSR     HCR_EL2, X0

```

```
MOV    X0, #0b10011    // DAIF=0000
MSR     SPSR_EL2, X0    // M[4:0]=10011 EL1 is SVC mode must match HCR_EL2.RW.

// Determine EL1 Execution state.
ADR     X0, e11_entry   // e11_entry points to the first instruction of SVC
MSR     ELR_EL2, X0     // mode code.

ERET

e11_entry:
// EL1 code here.
```
